



UE SPECTROSCOPIE AVANCÉE ET MATIÈRE DILUÉE

Basics of Photoelectron Spectroscopy

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Basic principles



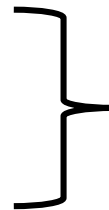
Photoelectron spectroscopy is based on Einstein's equation in photoelectric effect:

$$h\nu = I + \frac{1}{2}m_e v^2$$

I : Ionization energy for atoms/molecules in gas-phase, **UNIQUE** for each system

✓ $h\nu$ is known from the photon-source

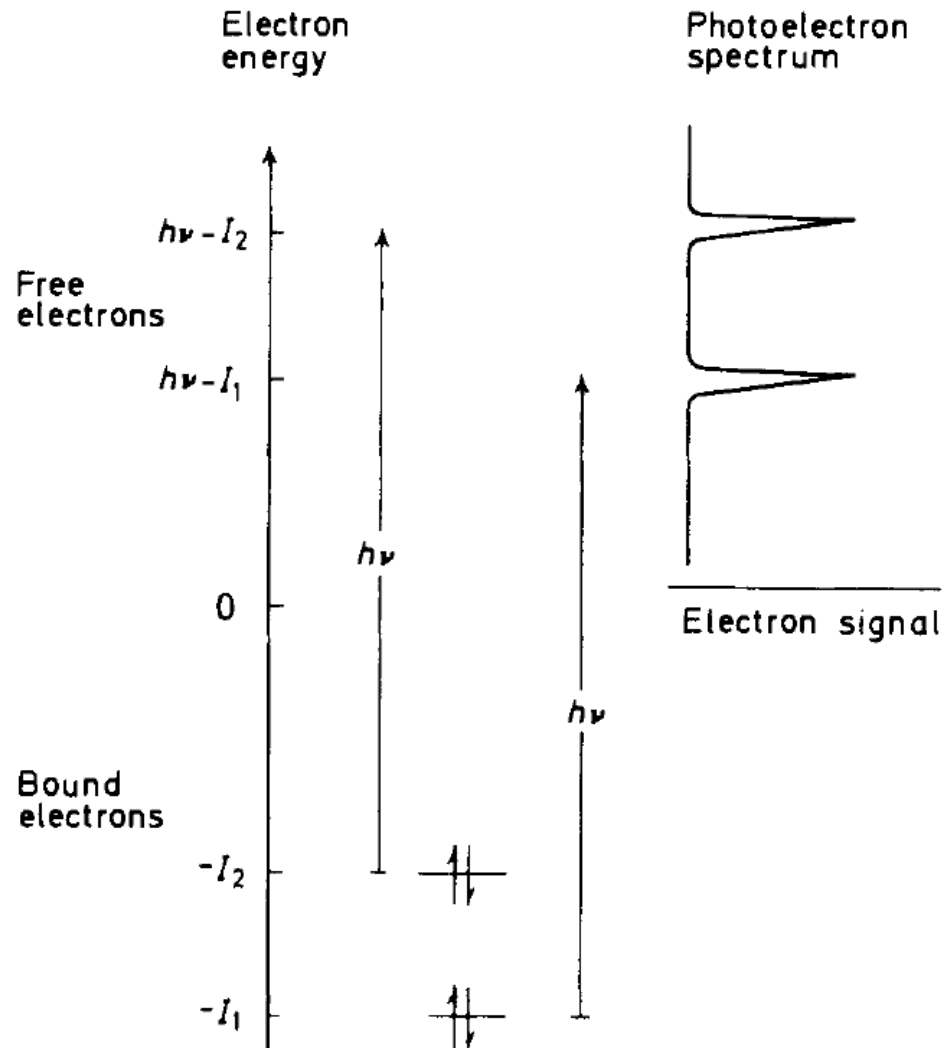
✓ $m_e v^2$ can be measured from spectrometer



I is experimentally determined

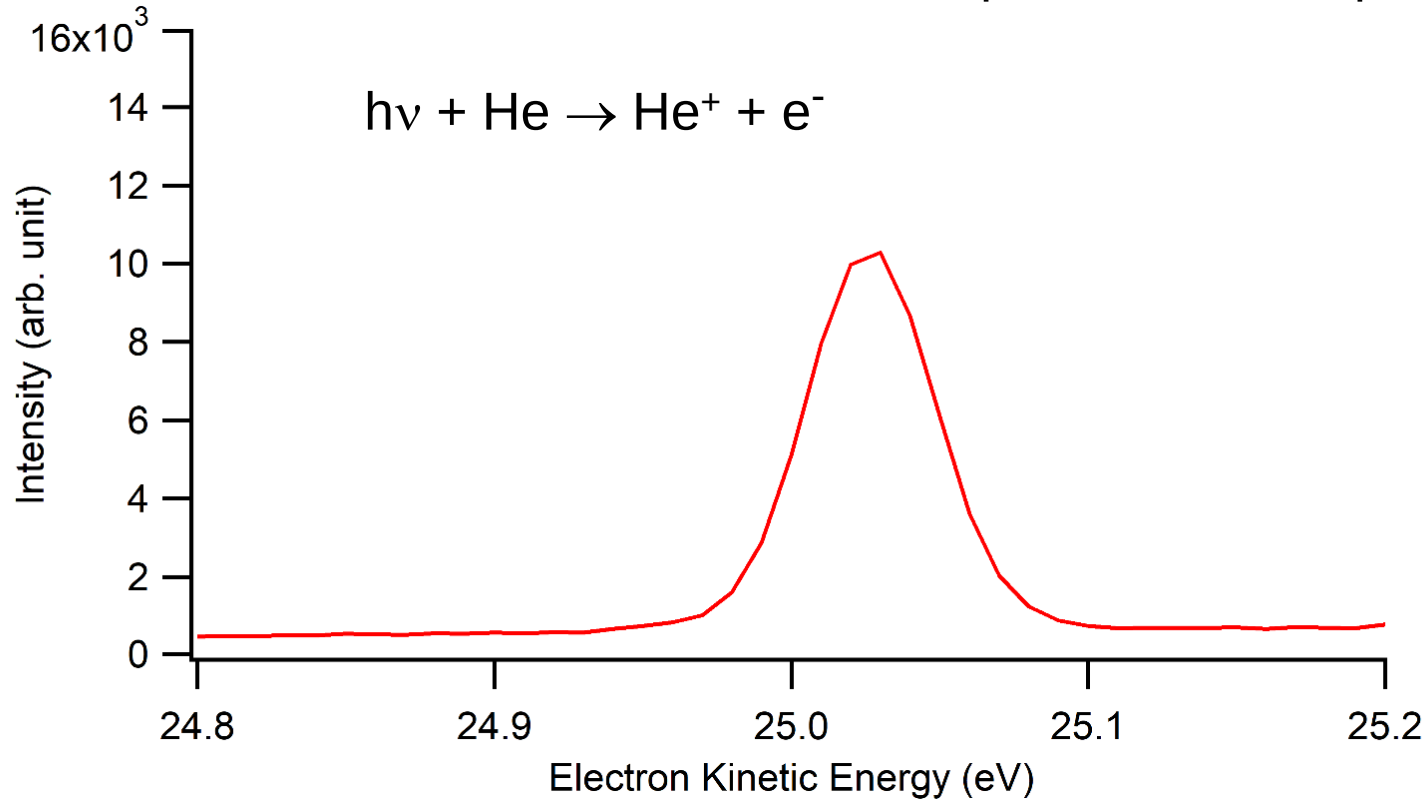
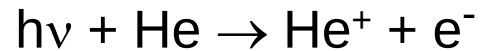
Detection of photoelectrons as a function of their kinetic energies → Photoelectron spectra

Basic principles



Atomic photoionization using XUV light

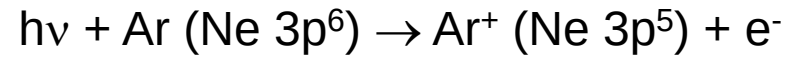
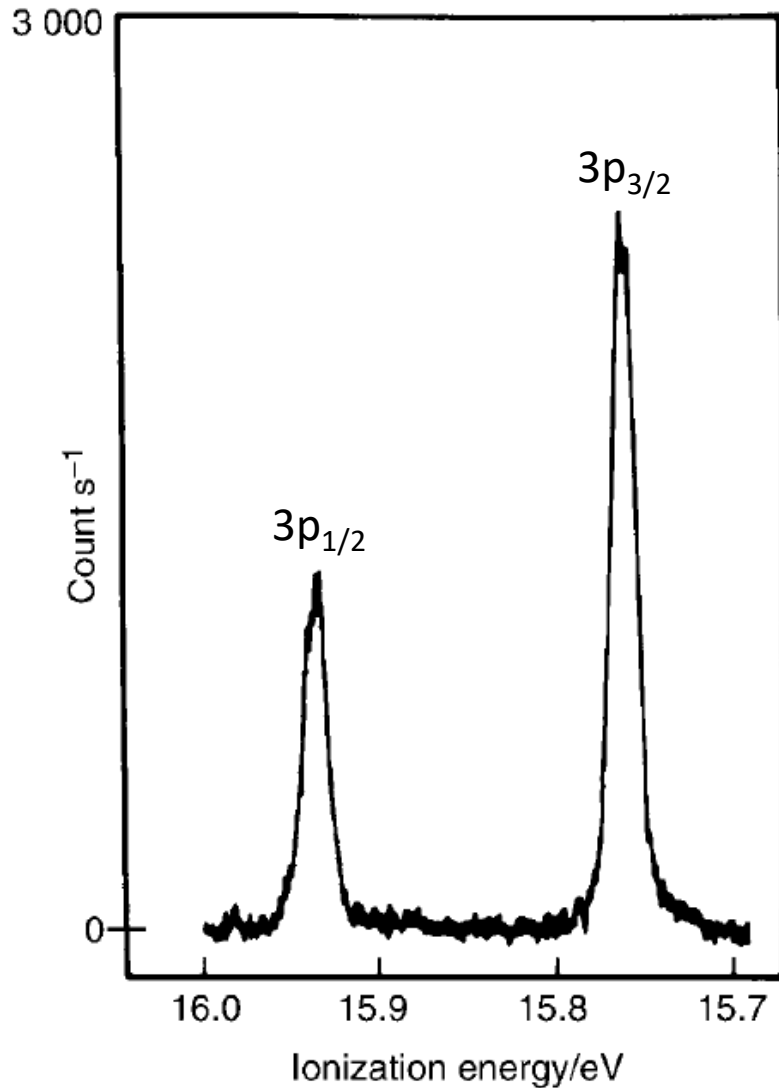
“Most well-known” atomic photoionization process:



Ionization potential of helium: 24.58 eV

What is the photon energy used here ?

Strength of photoelectron peaks

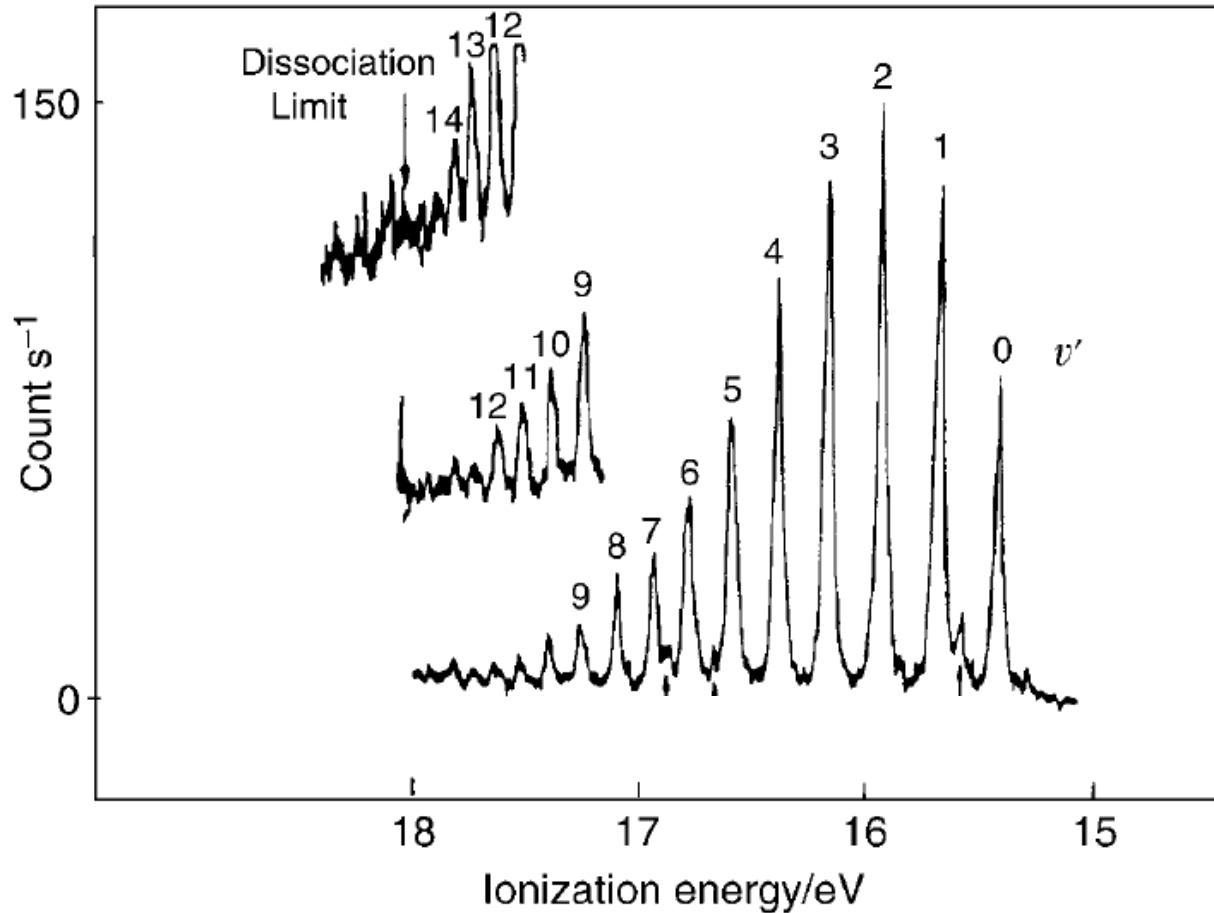


Notice the clear spin-orbit splitting of 178 meV.

The peak for 3p_{3/2} is twice more intense than that for 3p_{1/2}

Why ??

Molecular photoionization using XUV wavelength



Not just one peak, but multiple
peaks → Different vibrational levels

Koopmans' theorem

Each ionization energy is approximately equal in magnitude to an orbital energy calculated using a self-consistent field (SCF) :

$$I_j \approx -\epsilon_j$$

This may seem obvious, but it is not true always...

For example **electronic correlation**:

Electrons can 'talk' with each other via Coulomb interaction → Movement of one electron influences the others

Total Hamiltonian

$$\hat{H} = -\frac{\hbar^2}{2} \left(\sum_i \frac{\nabla_i^2}{m_e} + \sum_I \frac{\nabla_I^2}{M_I} \right) - \frac{1}{4\pi\epsilon_0} \left(\sum_{I,i} \frac{Z_I e^2}{|\vec{R}_I - \vec{r}_i|} - \sum_{i \neq j} \frac{e^2}{|\vec{r}_i - \vec{r}_j|} \right)$$

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Born-Oppenheimer
approximation

Koopmans' theorem

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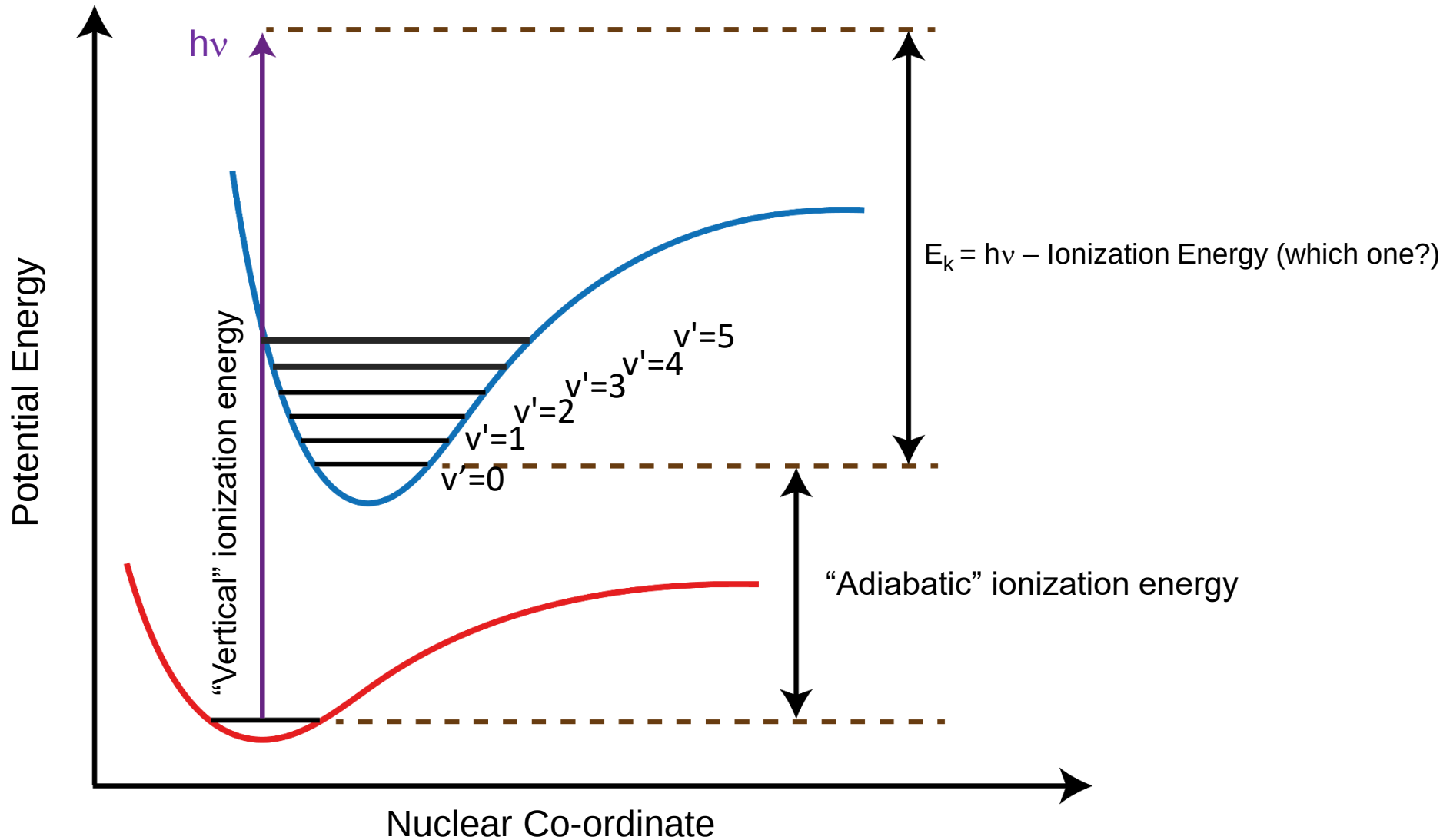
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Born-Oppenheimer approximation

$\approx \sum_i V(r_i)$ (SCF approximation)

Koopmans theorem + photoelectron spectra → Determination of molecular orbital picture

Quick survey of Franck-Condon theorem



Quick survey of Franck-Condon theorem

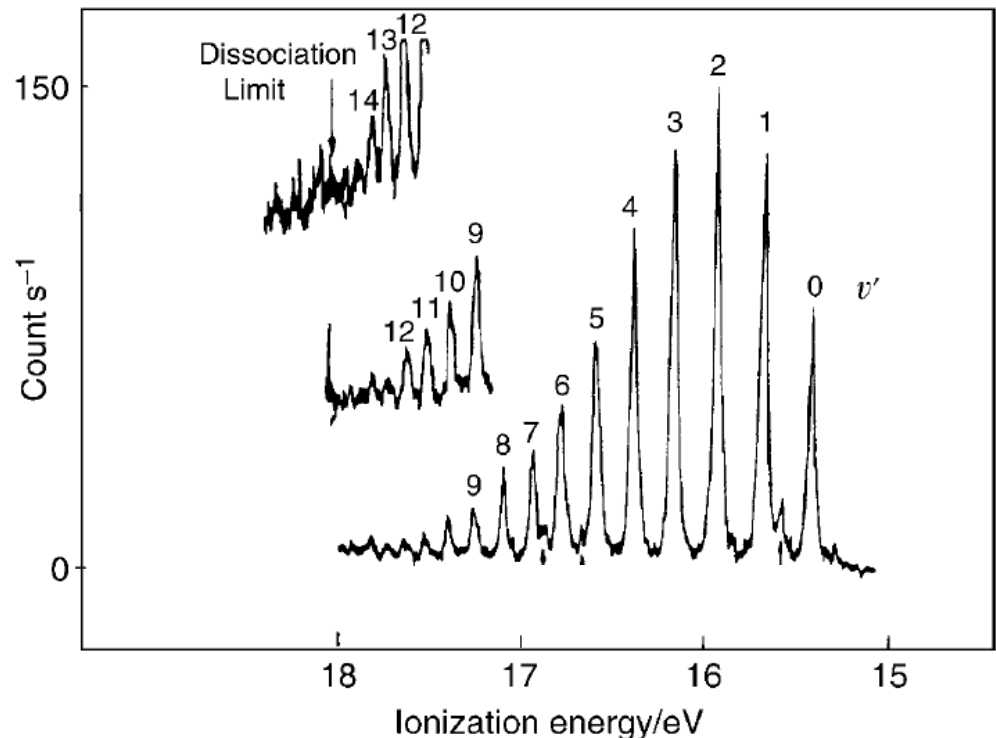
Intensities of the vibrational peaks is determined by the Franck-Condon factor which is the square of the overlap integral between two vibrational wave-functions:

$$\mathcal{F}_{v,v'} = \left| \int \chi_v \chi_{v'} dR \right|^2$$

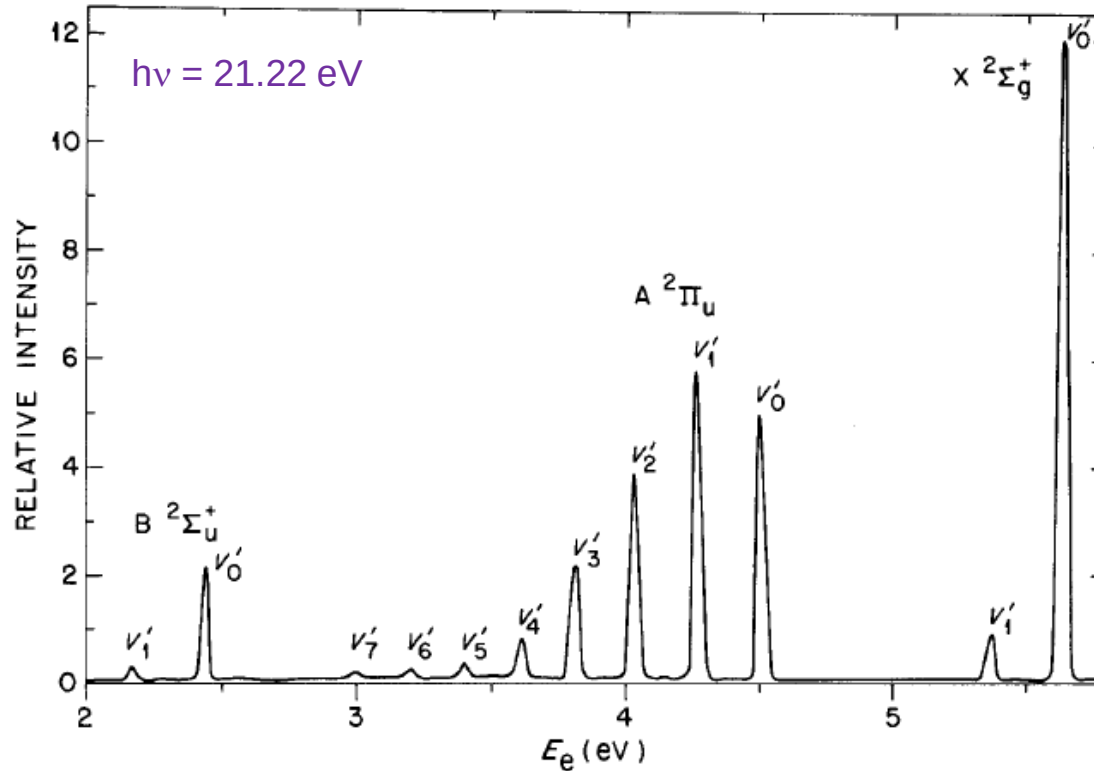
$$\sigma_v = \mathcal{F}_{v,v'} \sigma_{v'}$$

For H_2 , the transition probability is maximum for $v'=2$ and NOT $v'=0$.

Koopmans' theorem usually refers to the vertical ionization energy.



Photoionization of N₂ molecule using XUV light



For N₂

Two electrons in Highest occupied molecular orbital (HOMO)

Four electrons in HOMO-1

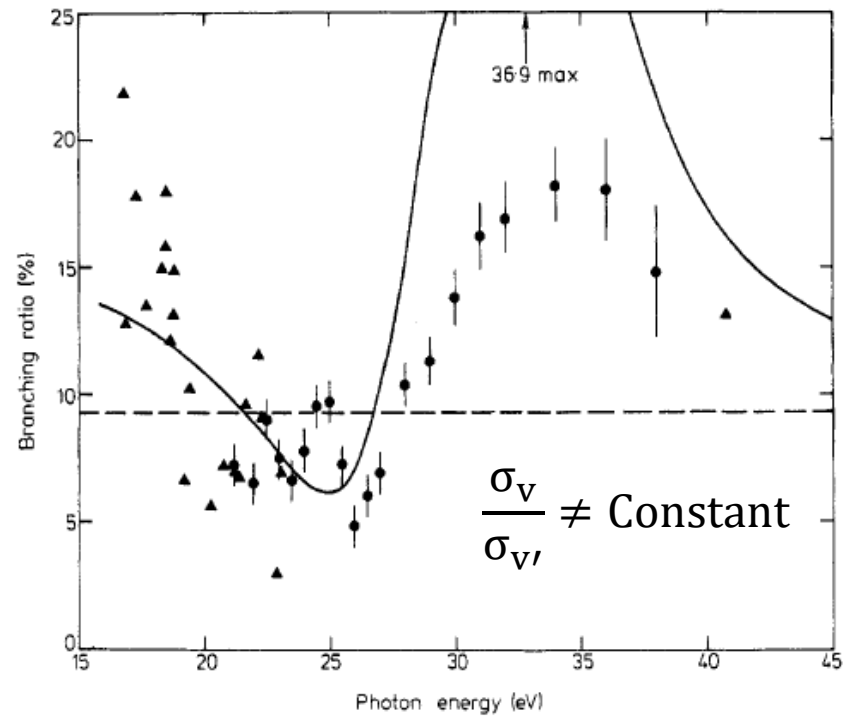
Two electrons in HOMO-2

...

Photoionization of N₂ molecule : shape resonance

$$\mathcal{F}_{v,v'} = \left| \int \chi_v \chi_{v'} dR \right|^2$$

$$\sigma_v = \mathcal{F}_{v,v'} \sigma_{v'}$$



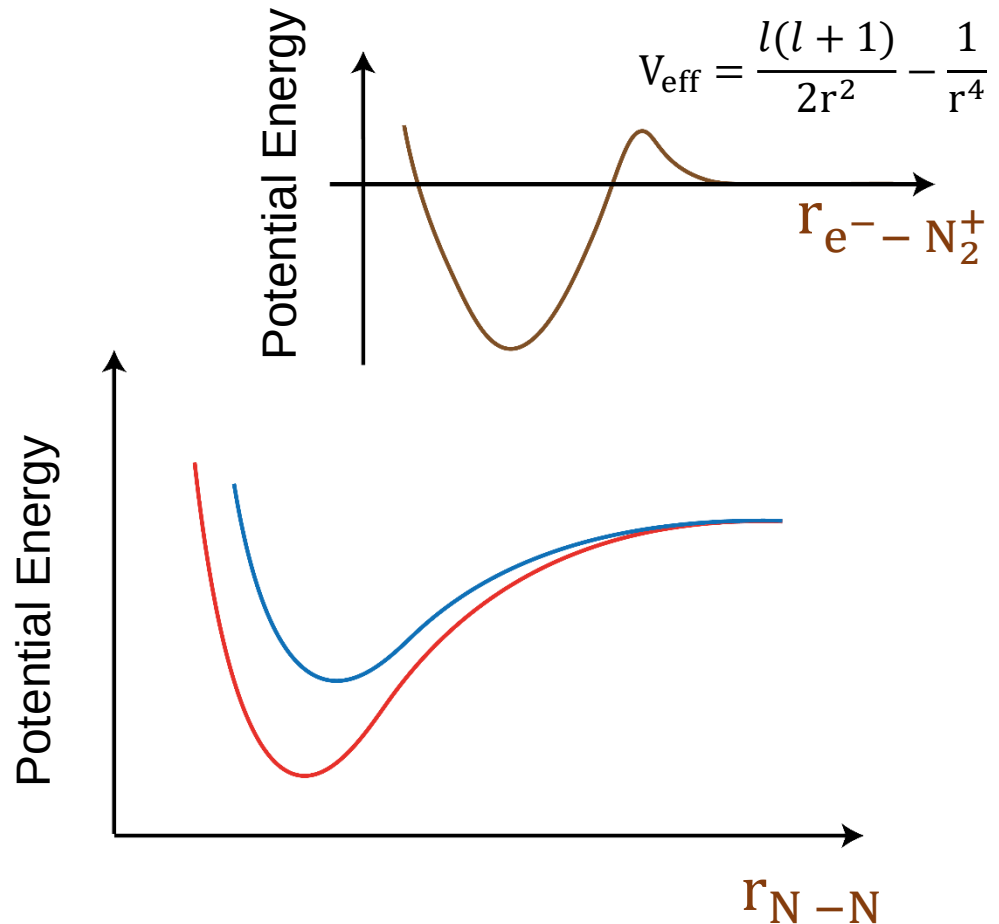
Photoionization of N₂ molecule : shape resonance

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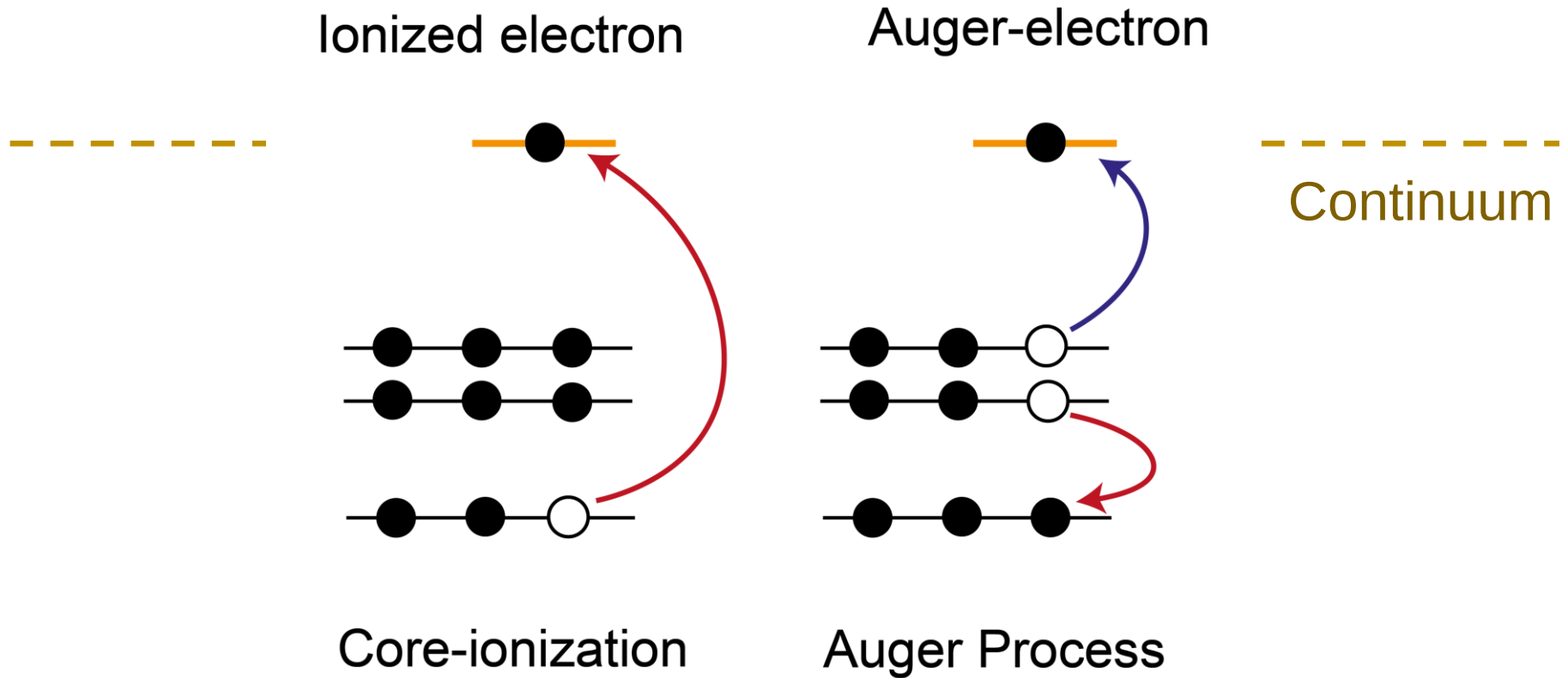
$$\sigma_v = \mathcal{F}_{v,v'} \sigma_{v'}$$

$$\frac{\sigma_v}{\sigma_{v'}} \neq \text{Constant}$$

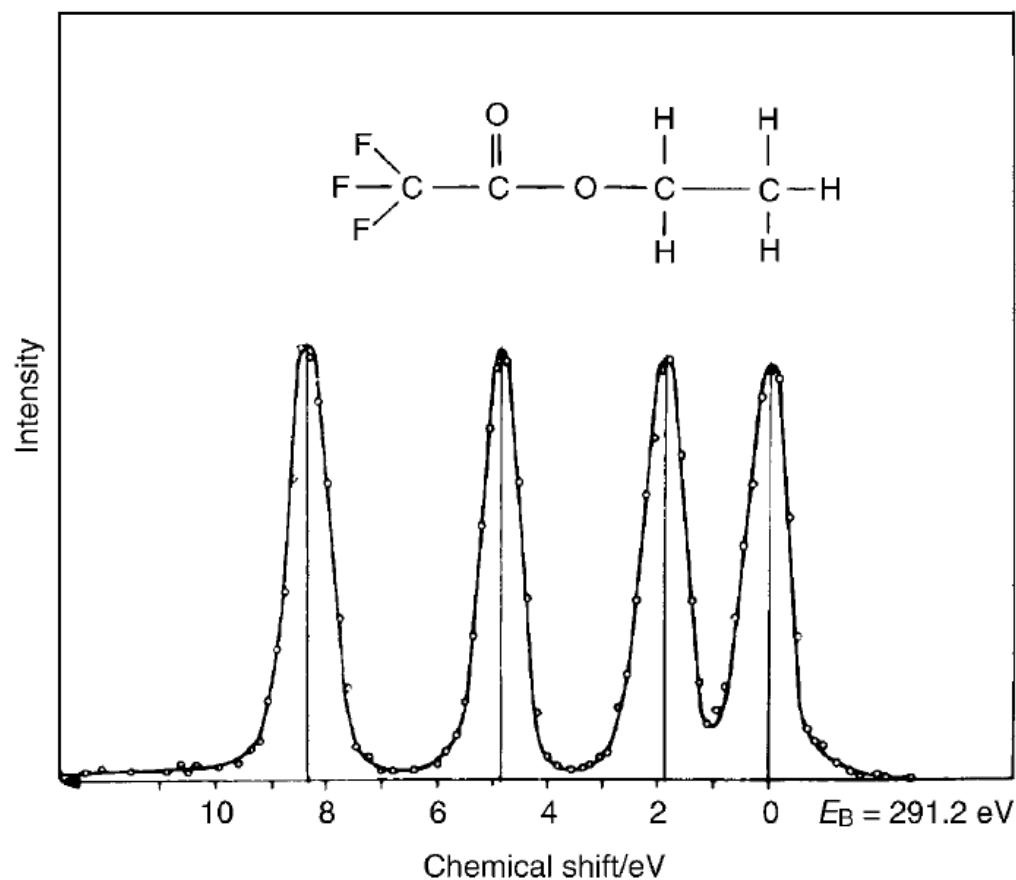
Shape resonance: Trapping by centrifugal **barrier**



Photoionization at higher photon energies using X-rays



Chemical shifts in molecular photoionization

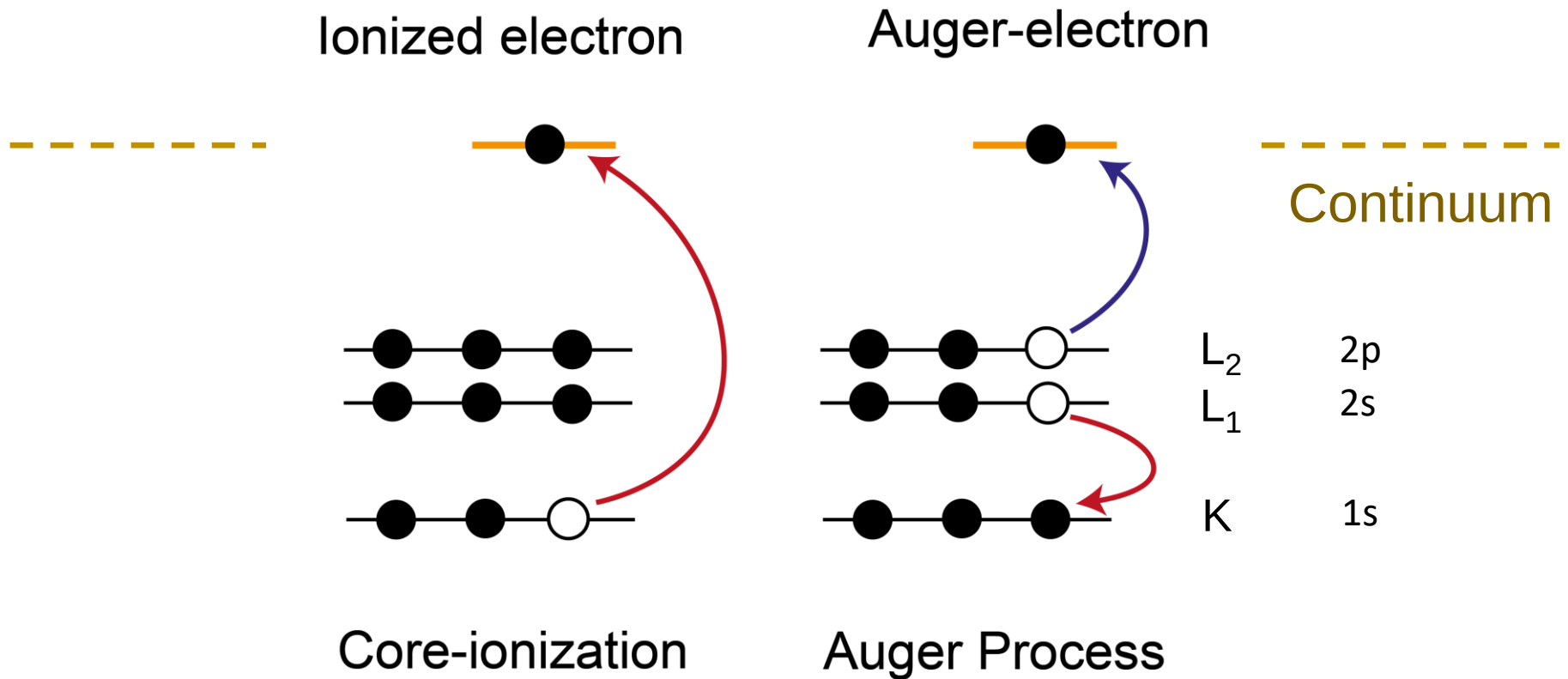


Chemical shift

$$\Delta E_{n\ell} = [E_{n\ell}(M^+) - E_{n\ell}(M)] - [E_{n\ell}(A^+) - E_{n\ell}(A)]$$

$$\Delta E_{n\ell} \approx -\epsilon_{n\ell}(M) + \epsilon_{n\ell}(A)$$

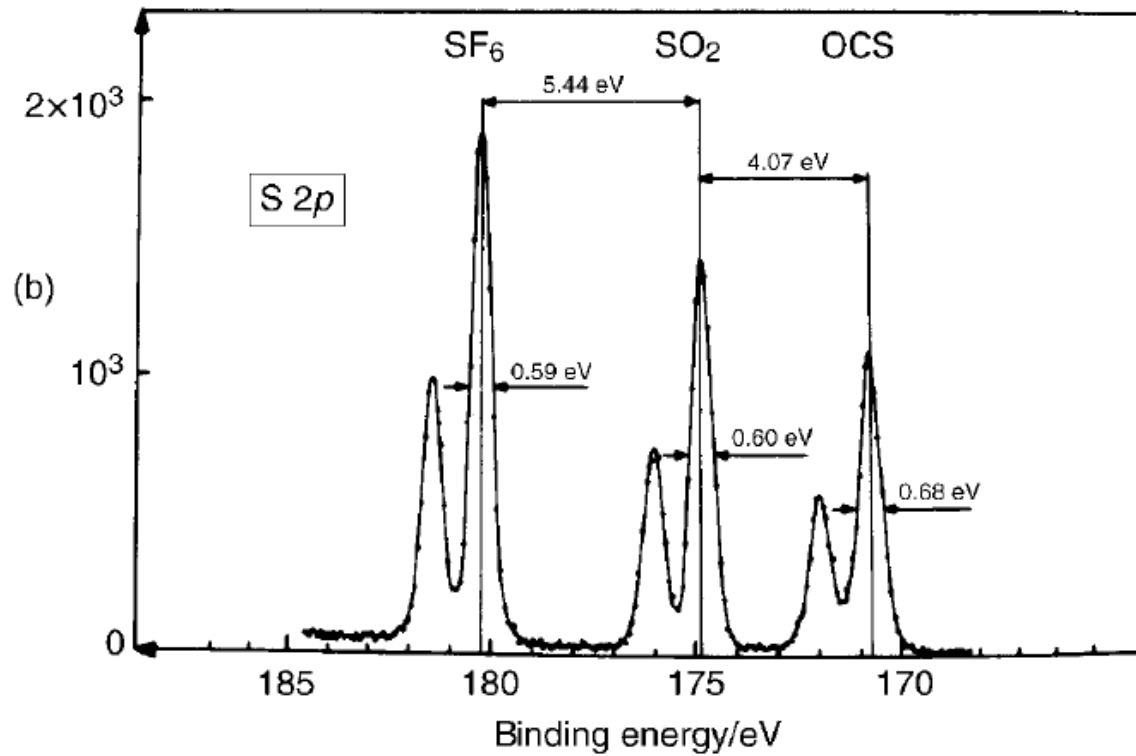
Auger electron spectroscopy



K-LL Auger electron energy:

$$E_K - E_{L_1} - E'_{L_2} = E_A$$

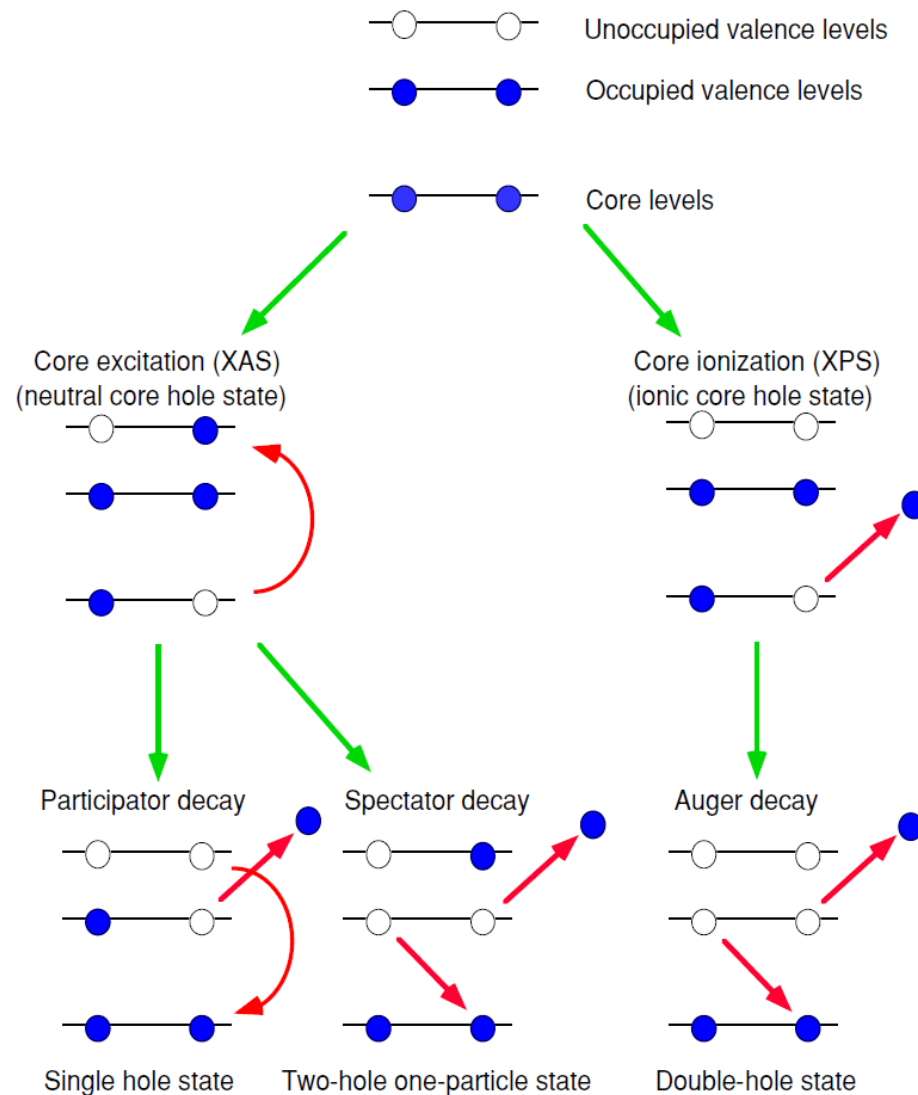
Auger electron spectroscopy



Example of Auger electron spectra

Resonant Auger electron spectroscopy

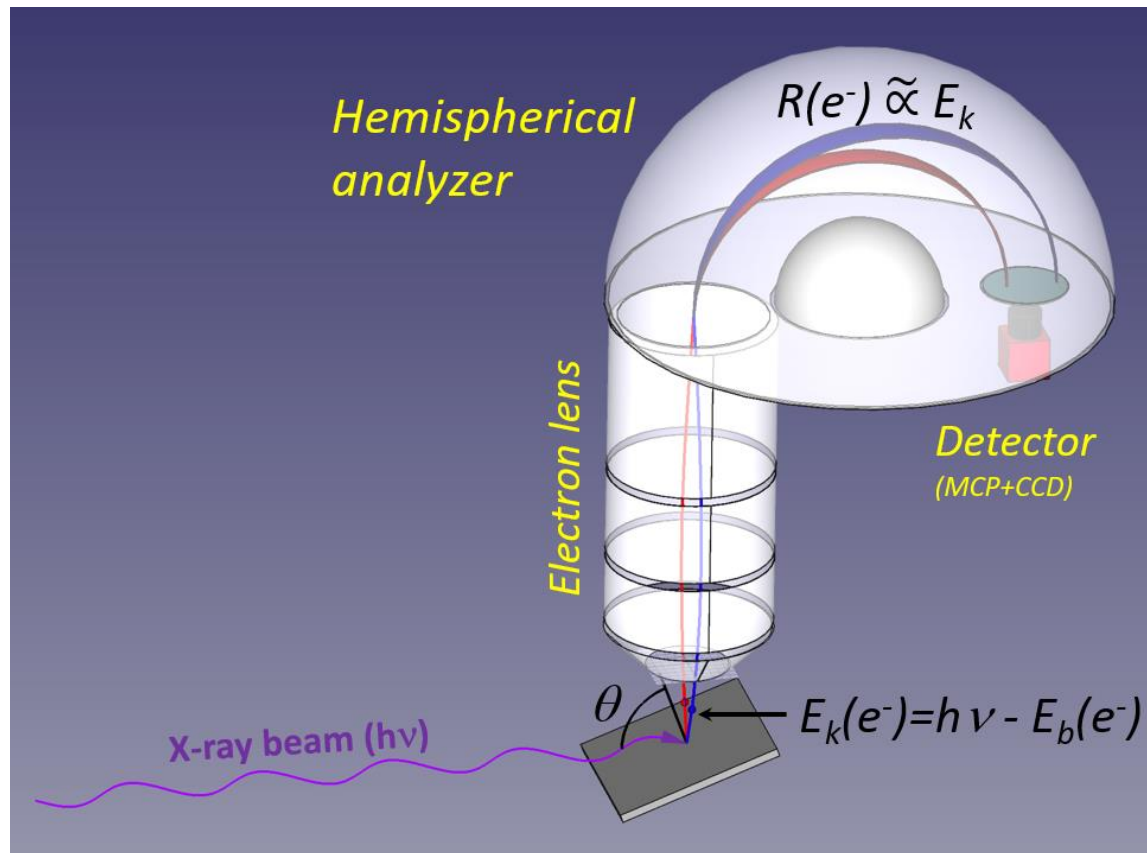
Core excitation, ionization and decay processes.



Experimental set-up

Three main ingredients:

1. **Sources** to get photons e.g., laser, synchrotrons, FELs
2. **Targets** to ionize e.g., **gas**, solids, liquids
3. **Spectrometer** to detect electrons



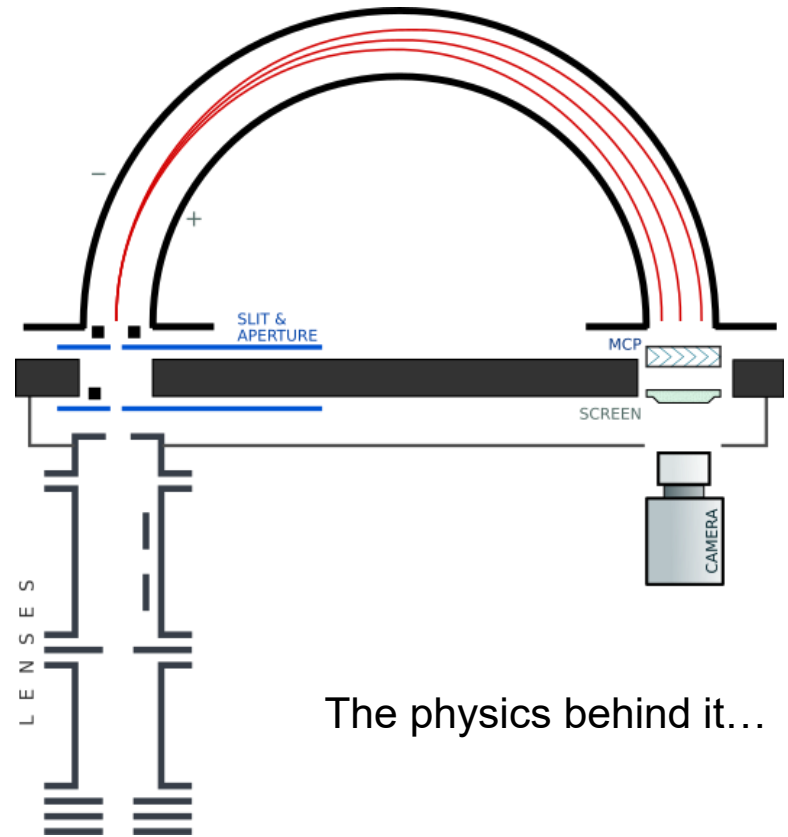
FlexPES beamline, MAX-IV, Lund, Sweden

Brief survey of hemispherical electron spectrometers



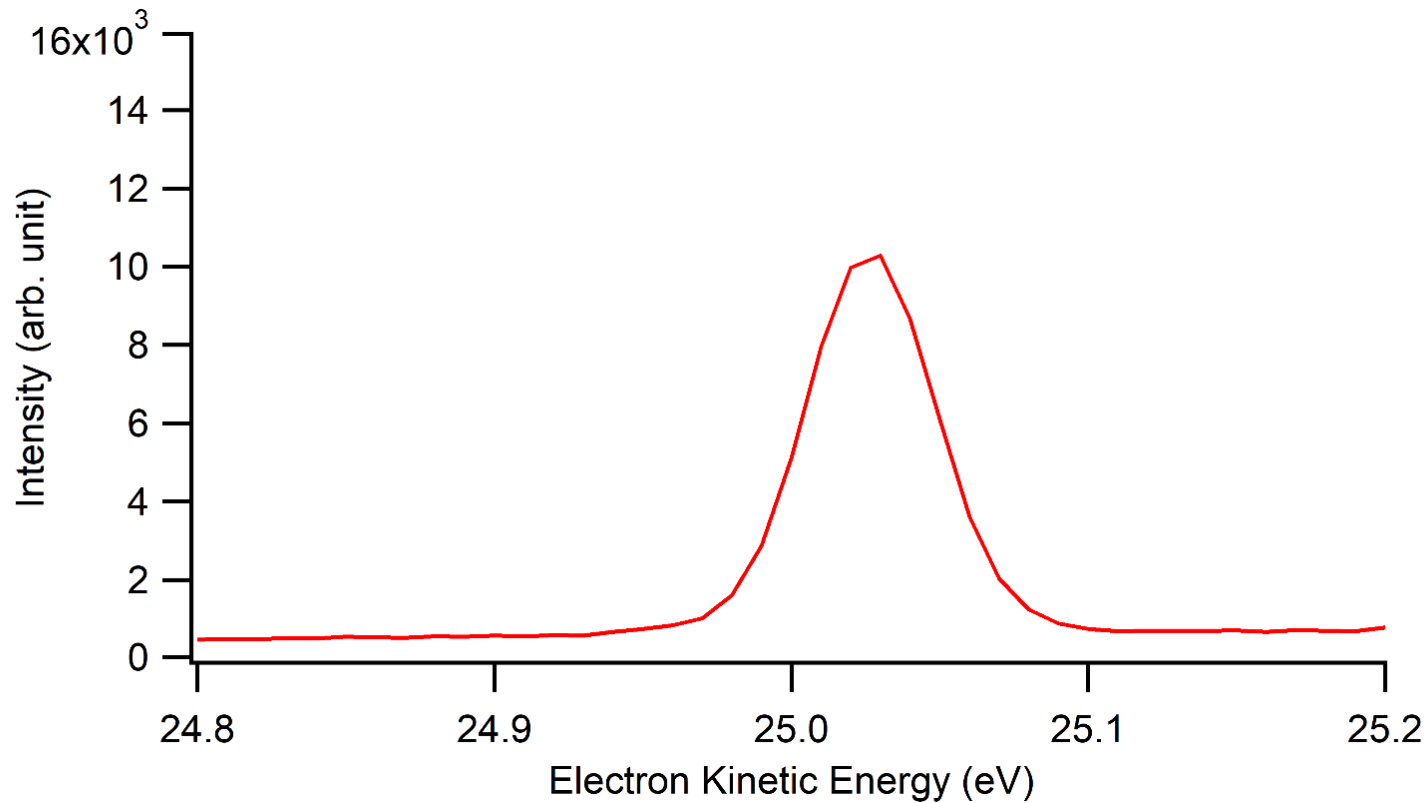
In reality

Applied electric field 'disperses' the electrons according to their kinetic energy.



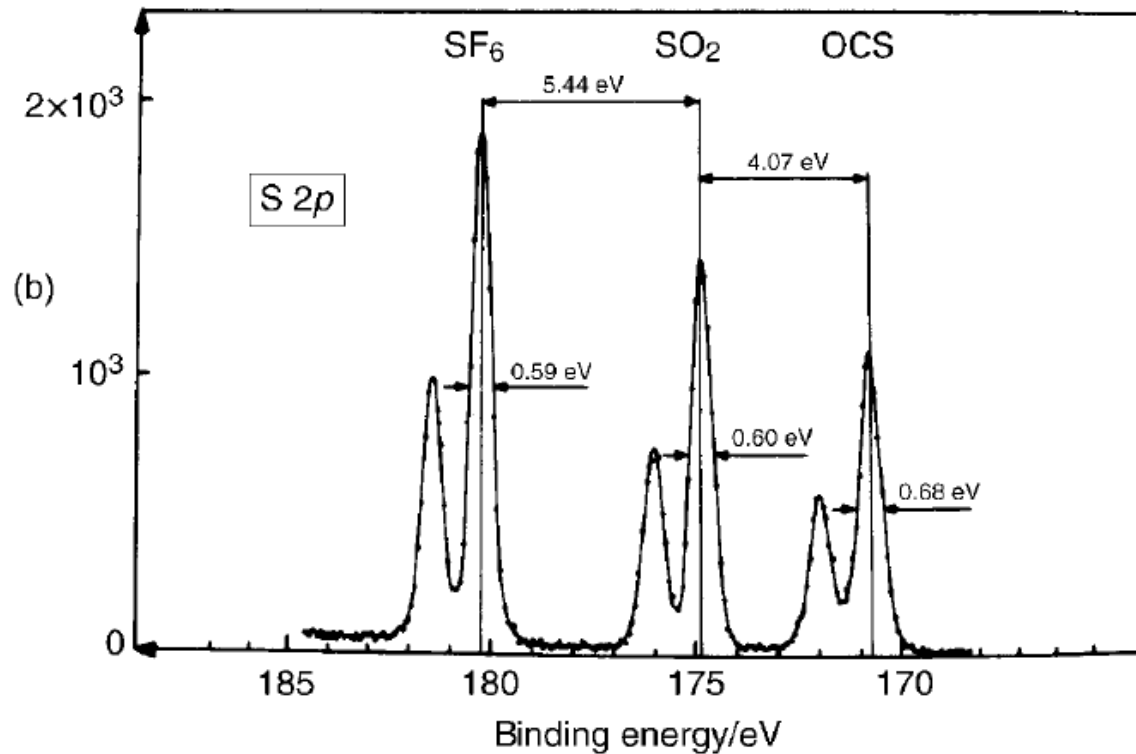
The physics behind it...

Width of photoelectron peaks



Depends on : i) Energy resolution of the photon pulse and ii) Resolution of the spectrometer

Width of Auger-electron peaks



Depends only on the resolution of the spectrometer. Why??